

Lecture 3 Homework

<http://thegrantlab.org/>

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To complete this homework you must be working on your own copy in **Google Docs**. Once you have filled in your answers to **Q1-Q4** in the space provided click **File > Download > PDF document** and upload to **gradescope** (link can be found on the class website).

Q1. [4pts] Consider the following multiple alignment of Transcription Factor Binding site DNA sequences

	1	2	3	4	5
Sequence 1	-	G	A	G	C
Sequence 2	C	T	A	G	A
Sequence 3	C	G	A	-	A
Sequence 4	A	G	C	G	A

Give the average profile (frequency matrix) of the above alignment by filling out the table below. The first position of the first column (i.e. position in the alignment) has been done for you, now complete the rest. You will use this table for answering questions 2 and 3 below.

	1	2	3	4	5
A	0.25	0	0.75	0	0.75
C	0.5	0	0.25	0	0.25
T	0	0.25	0	0	0
G	0	0.75	0	0.75	0
-	0.25	0	0	0.25	0

Q2. [2pts] What is the highest scoring sequence match to your profile above (question 1) and what is its score?

Sequence: **CGAGA**

Score: **3.5 -> 0.5+0.75+0.75+0.75+0.75= 3.5**

Q3. [2pts] Using your completed profile table above (from question 1) score the following two sequences (**S1** and **S2**):

S1. CTGGC Score: 1.75

S2. AGAGA Score: 3.25

Q4. [2pts] Following the heuristic threshold for a positive match proposed in Harbison et al. [Nature (2004) 431:99-104.] namely using the threshold for a positive match = 60% x Max Score
Are either of the two sequences in question 3 potential transcription factor binding sites? If so, why?

Threshold for Pos. Match: $(60\%) \times (Max\ Score) = (0.6) \times (3.5) = 2.1$

The threshold for a positive match is 2.1, calculated as 60% of the max sequence score. Any sequence with a score higher than this threshold qualifies as a potential transcription factor binding site.

S2 is a potential transcription factor binding site because it exceeds the threshold with an overall sequence score of 3.25; **3.25 > 2.1**. On the other hand, sequence 1 only has a score of 1.75 which fails to meet the threshold value of 2.1 and therefore cannot be considered a potential transcription factor binding site.